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The Economy Is Being Rewired Around Compute and Energy “And that’s a very different world.”

Jordi Visser

JVisser@22VResearch.com

22vresearch.com

Recently, I listened to a podcast with Raoul Pal and Julien Bittel on the macro world today titled *Why the Global Economy Is Changing Rapidly*. Near the end of the conversation, they talked about something I have written about in bits and pieces over the last couple of years. Because of the “this versus that” nature of the discussion, combined with all the recent growing bubble talk on X, I thought it was important to share my thoughts on their conversation.

Early in the podcast, Raoul set the framework:

“I’ve had this whole thesis around this universal code that everything is funneling from energy into intelligence and it’s driving the geopolitical process. It’s driving the investment process. It’s driving literally everything right now.”

Later, he added:

“And you know, we’re going to shift from the magic formula of GDP equals population growth plus debt growth plus productivity growth into the economic singularity version, which is AI plus robots plus humans. And productivity is how much intelligence you can produce per unit of energy.”

Then Julien summed it up:

“Yeah. We’re leaving a world that’s been dependent on capital and labor and moving into a world that’s entirely dependent on compute and energy. And that’s a very different world.”

This gets to the heart of why we are watching something that has never happened before. Experienced people keep saying they have seen this AI playbook before and that it ends badly. I understand the instinct, because every major investment cycle eventually has excess, disappointment, and corrections. But this is also different. We are watching the industrial world change in front of our eyes, and with it the cycles most of our models and memories were built on.

To summarize what is happening as billions of digital employees, and soon humanoids, enter the workforce: we exited the old economic formula of capital, labor, population, debt, and productivity, and entered a new world where compute and energy become the foundation of growth, the base of the five-layer cake, because productivity is now measured by how much intelligence can be produced per unit of energy.

For most of modern economic history, the business cycle has been understood through the tension between labor and capital. Companies borrowed money, hired workers, built capacity, and expanded until demand slowed, margins compressed, asset prices fell, and the debt that financed the expansion became the problem. When the cycle turned, labor became the release valve. Companies cut workers to protect levered balance sheets, reduce costs, and repair the damage created by overinvestment. That was the textbook cycle: capital overreached, labor absorbed the shock, and credit transmitted the pain.

The AI cycle is following a different model. It is the reindustrialization of intelligence, and its constraints are increasingly compute and energy.

Jensen Huang has described AI as a five-layer cake: energy, chips, infrastructure, models, and applications. That framing matters because the bottom of the stack is physical. Before there are agents, copilots, digital workers, enterprise automation, robotics, or intelligence-as-a-service, there must be electricity, semiconductors, memory, networking, cooling, land, power interconnections, and data centers. AI may feel like software to the user, but economically it begins as industry. It starts with atoms before it becomes bits. Huang's other formulation is even simpler: the input is electrons, the output is tokens. That is the new production function. The factory consumes energy and produces intelligence.

This is also why the AI cycle is so different from the software cycle that preceded it. The irony is that the great software winners already showed us that the labor-capital relationship was breaking. It is what drove me from spending a month of the year in China to visiting Singularity University more than a decade ago to update my macro framework to be based on exponential innovation. Microsoft, Google, Meta, Amazon, and other platform companies scaled revenue, margins, and market value without scaling labor or brick-and-mortar locations in the old industrial way. The opposite of ghost cities. Revenue per employee and profit margins for the leading technology companies moved into numbers that would have been unimaginable in traditional manufacturing, retail, or transportation.

The software world proved that once a platform is built, additional revenue can arrive with very little incremental headcount. That was the first crack in the old model. It is one of the reasons why, since the Great Financial Crisis, there has not been a credit and labor led recession. The only recession: the government-driven Covid shutdown. AI widens this crack even further. If software allowed companies to scale with fewer people, AI allows companies to scale with a digital labor force that works 24 hours a day, 7 days a week, 365 days a year. The age of relying on historical economic analogies is breaking, and so are the memories and patterns investors associate with bubbles and investment cycles.

Tokens are the food source of these digital workers. A human employee consumes wages, benefits, office space, management time, and training. Then they have families, buy cars, have children, and leverage up through a home. A digital employee consumes compute, memory, electricity, and data. That changes the bottleneck, and it also changes the business cycle. In the old economy, a company needed more workers to produce more output. In the AI economy, a company needs more tokens. More tokens require more compute. More compute requires more energy. Labor still matters, but the marginal unit of growth is changing. The next unit of productivity is less likely to be a newly hired person and more likely to be a software agent running on infrastructure somewhere in Virginia, Texas, Arizona, Ohio, Taiwan, or Ireland. For human employees, the dream of the rising the corporate ladder has turned into the anxiety of AI driven job losses. This leaves the stock market at all-time highs and consumer sentiment suggesting we are in a recession.

That is why today's AI capital spending should not be analyzed like a normal credit bubble. In prior cycles, excess borrowing funded excess capacity, and when demand disappointed, the debt remained. Investors are understandably focused on the size of the debt being raised, the amount of free cash flow being consumed, and whether the capex can earn an adequate return. Those questions matter. But in the old cycle, the first place we looked was debt-to-equity, refinancing risk, and whether companies had borrowed beyond the capacity of their balance sheets. This cycle looks different. These companies have the ability to raise hundreds of billions of dollars precisely because their balance sheets, income statements, market positions, and cash flows give them access to capital on a scale that highly levered industrial borrowers in prior cycles never had. The hyperscalers are spending hundreds of billions of dollars on AI infrastructure, but they are highly profitable businesses with dominant market positions, enormous cash flows, and balance sheets that look nothing like the overextended borrowers of prior investment booms.

They also have over \$1 trillion, and growing, in cloud receivables for the world's digital workforce. Demand is already far greater than supply. Demand grows daily in the form of bits, while compute is dependent on the complexity of atoms. Every dollar of AI capex still has to earn an adequate return, and some of it will disappoint. The real risk is that investors keep using the wrong historical template.

What is also different is that many of the companies building the infrastructure and intelligence stack are increasingly coming to public markets to raise capital, and investors continue to absorb the supply. Google recently came to the market and raised an astonishing \$85 billion in equity, with Berkshire Hathaway purchasing roughly \$10 billion despite the stock having doubled over the prior year. Meta has leaked its desire to tap the public equity markets. Behind Google and Meta sits an extraordinary pipeline of potential issuers. OpenAI, Anthropic, and SpaceX alone could eventually bring more than \$4 trillion of combined market capitalization to public markets, with the money funding even more capex.

In a traditional cycle, investors might struggle to finance that scale of issuance. Instead, capital continues to flow toward the companies building the compute, energy, infrastructure, and intelligence layers of the AI economy. All of this supply is arriving while the broader stock market continues to move higher, a sign of how powerful the demand for ownership in the AI buildout has become.

At the same time, the household transmission mechanism has also changed. Despite how often you hear that very few people benefit from the rising stock market, the facts are that roughly two-thirds of Americans now own stocks directly or indirectly, most often through retirement accounts and pension funds. That means a rising stock market does more than enrich a narrow owner class; it also supports household confidence for short term expenditures, retirement balances, and consumption, even if the savings rate goes down.

On the other side of the distribution, government transfers have become a much larger share of personal income than they were fifty years ago. The result is a strange form of economic stabilization also caused by exponential innovation through its Schumpeterian process. Asset appreciation supports one large part of the population, while transfer payments support another. This does not solve wealth concentration or inequality, and it may even worsen resentment, but it helps explain why the economy can keep absorbing shocks without falling into the old recession pattern. Unless there is a large wave of layoffs from rapidly declining profit margins, the traditional business-cycle trigger is harder to find.

The better place to look for a supply demand cycle and mismatch is the AI adoption cycle colliding with a physical bottleneck. Human adoption and enterprise adoption are still at the very early stage. ChatGPT reaching massive global usage in only a few years showed how fast the demand side can move when intelligence becomes both a consumer and enterprise interface. Anthropic's recent parabolic rise in ARR confirms it. But enterprise AI is still uneven. Many companies are being forced to accelerate plans because of competition. They are experimenting, and fewer have rewired workflows. Only a minority have scaled AI deeply across their organizations.

That means the demand for intelligence, along with the capex cycle, is still in the early innings even as the supply chain already looks strained. This is the heart of the new cycle: bits can compound almost instantly, but atoms cannot. Models, apps, agents, and demand can grow exponentially. Power plants, transmission lines, transformers, fabs, substations, and data centers cannot.

This is where we stand today. The bottlenecks and shortages across the physical AI supply chain are not easing. They are spreading. Last year, the conversation was mostly about transformers, grid connections, gas turbines, and power availability. At the time, memory still looked relatively abundant. Now the shortages have moved deeper into the compute stack. We are hearing about tighter supply in memory, CPUs, advanced packaging, networking equipment, and the power components needed to support higher-density racks.

That matters because it shows this is not one isolated constraint. It is a rolling shortage cycle across the entire physical layer of AI. As soon as one bottleneck is identified, demand pushes into the next constraint. The market may reprice the winners and losers along the way, but the broader message is clear: the digital side of AI can scale demand faster than the industrial side can scale supply. Then if you add in the back-to-back years of commodity and mineral shortages due to rare earth trade wars and the shutting of the Strait of Hormuz you have more warning signs around the supply chain. Petrochemicals, silver and copper are also necessary for the AI buildout and are all under strains as well.

AI does not eliminate corrections, risk, overinvestment, or disappointment. There will still be investment corrections. There will still be moments when expectations run ahead of reality. There will still be periods when markets have to reprice the timing, profitability, and scale of the buildout. But in the AI cycle, I believe those corrections will be less about the old credit-driven relationship between labor and capital and more about bottlenecks and shortages in the physical world.

The demand side is being driven by digital employees that can work continuously and consume more intelligence every day. The supply side is constrained by atoms: power plants, chips, memory, transformers, cooling systems, transmission lines, and data centers. That mismatch is the new cycle. The risk is real, but it is different. It is the risk that hypergrowth in digital intelligence runs faster than the physical world can feed it. During those times, expect to see major volatility in the AI infrastructure trades as investors lose clarity on when the earnings and ROIC can come through relative to the speed and growth of the capex spend.

That is the reality of the cycle we have entered and where we are likely to be for a long time. The old cycle asked whether labor costs were rising too fast, whether credit was too loose, whether companies had borrowed too much, and whether demand would fall enough to force layoffs. It was built around capital versus human labor. The AI cycle asks different questions. How do we feed the digital employees? Is there enough power? Are there enough chips? Is there enough memory? Can data centers get connected to the grid? Can the physical infrastructure keep pace with the demand for intelligence? Can enterprises adopt AI fast enough to justify the buildout, and can the buildout happen fast enough to satisfy adoption?

There are no textbooks for this because the economy has never reindustrialized intelligence before. The business cycle is no longer simply labor versus capital. Increasingly, it is compute versus energy and as Raoul and Julien said, “And that’s a very different world.”

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